

Comparative Analysis of CNN-Based and Vision Transformer-Based Image Caption Generator Using Transformer Models

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Abstract—Automatic image captioning requires the joint modeling of visual content and natural language, making the choice of visual encoder a critical architectural decision. This paper presents a controlled comparative analysis of two encoder-decoder image captioning systems that share an identical Transformer decoder but differ in their visual encoder: one employs a pretrained ResNet-50 convolutional neural network and the other uses a pretrained Vision Transformer (ViT-B/16). Both models are trained and evaluated on the Flickr30k benchmark dataset using standard caption quality metrics, including BLEU-1 through BLEU-4, METEOR, ROUGE-L, and CIDEr. The ViT-based model attains slightly higher BLEU-1 through BLEU-3 scores, but the ResNet-based model achieves stronger BLEU-4 (0.2074 vs. 0.2045), METEOR (0.1993 vs. 0.1984), ROUGE-L (0.4951 vs. 0.4914), and CIDEr (0.4263 vs. 0.4137), indicating better long-form n-gram consistency and overall caption consensus. In terms of efficiency, the ResNet-based model reaches convergence in 216.31 minutes over 24 epochs, compared with 500.44 minutes over 24 epochs for ViT, representing a $2.31\times$ reduction in training time. Inference latency is comparable for both models (41.06 vs. 41.35 ms per image). These results indicate that under the present training setup and dataset scale, a CNN encoder provides a stronger quality-efficiency trade-off than a ViT encoder.

Index Terms—image captioning, ResNet-50, Vision Transformer, ViT-B/16, Transformer decoder, Flickr30k, BLEU, CIDEr, encoder-decoder, multimodal learning

I. INTRODUCTION

Image captioning is a challenging multimodal task that aims to automatically generate natural language descriptions for images by combining techniques from Computer Vision (CV) and Natural Language Processing (NLP). Unlike traditional image classification or object detection, image captioning requires a system to understand not only the objects present in an image but also their relationships, attributes, and contextual information in order to produce coherent and grammatically correct sentences. The ability to automatically describe visual scenes has become increasingly important due to its applications in assistive technologies for visually impaired individuals, intelligent image retrieval, autonomous vehicles, surveillance systems, robotics, social media content analysis, and human-computer interaction.

Early image captioning systems relied on handcrafted visual features combined with template-based or retrieval-based sentence generation. The rapid advancement of deep learning introduced encoder-decoder architectures inspired by neural machine translation, in which Convolutional Neural Networks (CNNs) encode visual representations and Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, generate captions sequentially.

Despite remarkable improvements, RNN-based decoders suffer from difficulties in capturing long-range dependencies and cannot be fully parallelized during training. The Transformer architecture addressed these limitations by replacing recurrence with self-attention mechanisms that enable efficient modeling of global dependencies. Transformer-based decoders have since become the dominant approach in modern image captioning systems.

CNN architectures such as ResNet remain highly effective at learning hierarchical local features and are widely adopted as image encoders due to strong inductive biases for translation equivariance and locality. Vision Transformers (ViTs) have emerged as a powerful alternative by applying self-attention directly to sequences of image patches, enabling explicit modeling of long-range spatial dependencies. While ViTs have demonstrated state-of-the-art performance across many vision tasks, they generally require larger datasets and more computational resources than CNN-based models.

This study presents a controlled comparative analysis of two image captioning architectures employing a common Transformer decoder while differing only in their visual encoder. The first architecture uses a pretrained ResNet-50 encoder; the second employs a pretrained ViT-B/16 encoder. Using the Flickr30k benchmark dataset, both models are evaluated on standard caption quality metrics including BLEU-4, METEOR, ROUGE-L, and CIDEr, along with training time, convergence behavior, and inference latency. By isolating the effect of the visual encoder, this work provides a comprehensive assessment of the trade-offs between CNN-based local feature extraction and Vision Transformer-based global contextual representation.

The major contributions of this paper are:

- A comparative implementation of ResNet-50 and ViT-B/16 visual encoders paired with an identical Transformer decoder under matched training conditions.
- Quantitative evaluation on Flickr30k using BLEU-1–4, METEOR, ROUGE-L, and CIDEr metrics on 1,000 test images.
- Comparative analysis of computational efficiency, including encoder size, convergence rate, total training time, and inference latency.
- A discussion of the practical trade-offs between CNN local feature extraction and ViT global contextual modeling for image captioning.

II. LITERATURE REVIEW

Image captioning integrates computer vision and natural language processing to automatically generate coherent textual descriptions of visual content [1]. The field has evolved from early encoder-decoder models through attention-augmented architectures to recent vision-language foundation models [2].

Vinyals *et al.* introduced the Neural Image Caption (NIC) model, establishing the encoder-decoder paradigm by coupling a pretrained GoogLeNet encoder with an LSTM decoder [3]. Xu *et al.* extended this framework with soft attention, enabling the decoder to focus on spatially relevant image regions at each generation step [4]. Anderson *et al.* further improved caption quality by incorporating bottom-up attention over Faster R-CNN object proposals, achieving state-of-the-art performance on MS COCO [5].

Vaswani *et al.* introduced the Transformer, replacing sequential recurrence with multi-head self-attention to enable fully parallelized training and better modeling of long-range dependencies [6]. Transformer decoders rapidly superseded LSTM-based generators in image captioning by producing more fluent and contextually consistent captions. He *et al.* introduced ResNet, addressing the degradation problem in very deep CNNs via residual skip connections, making ResNet-50 one of the most widely adopted visual encoders for captioning tasks [7].

A major paradigm shift occurred with the Vision Transformer (ViT) of Dosovitskiy *et al.*, which partitions images into fixed-size patches and applies Transformer self-attention across the patch sequence, enabling direct modeling of global contextual relationships [8]. ViTs demonstrated competitive or superior image classification performance while exhibiting strong scalability with data and model size.

More recently, large-scale vision-language pretraining has pushed captioning performance substantially higher. BLIP [9] introduced a bootstrapping strategy that unifies understanding and generation by filtering noisy web image-text pairs. BLIP-2 [10] extended this approach by bridging frozen image encoders and frozen large language models (LLMs) via a lightweight Querying Transformer (Q-Former), enabling highly parameter-efficient multimodal pretraining. Zhu *et al.* demonstrated that BLIP-2 can serve as a vision question-answering backbone

in an interactive captioning pipeline (ChatCaptioner), identifying 53% more objects than BLIP-2 alone and producing significantly more informative descriptions [11]. Despite the impressive performance of these Multimodal Large Language Models (MLLMs), hallucination remains a critical challenge: models can generate plausible-sounding but factually incorrect descriptions of visual content. Li *et al.* proposed the POPE benchmark specifically to evaluate object hallucination in large vision-language models, revealing that even state-of-the-art MLLMs systematically hallucinate objects not present in images [12]. These findings emphasize the need to understand how different visual encoders contribute to grounding accuracy in caption generation.

Although ViTs have demonstrated strong representational capabilities, relatively few studies have performed fully controlled comparisons in which only the visual encoder differs while the Transformer decoder, vocabulary, training strategy, dataset, and evaluation protocol remain identical. Such controlled comparisons are essential for isolating the contribution of the visual feature extraction mechanism. Specifically, because ViT relies entirely on learned patch embeddings without the locality and translation-equivariance inductive biases built into CNNs, it is expected to model global scene semantics more effectively but to be less data-efficient than ResNet on moderate-scale datasets such as Flickr30k. Quantifying this trade-off under a controlled experimental design constitutes the central research gap addressed by this study.

III. METHODOLOGY

Both captioning systems adopt an encoder-decoder framework in which a visual encoder extracts image representations and a shared Transformer decoder generates captions autoregressively. The two systems differ solely in the visual encoder; all other components are identical.

A. CNN-Based Captioning Model

The CNN encoder is a pretrained ResNet-50 [7] with the final classification layer removed. The spatial feature map from the penultimate global average pooling layer produces a 2,048-dimensional vector, which is projected to a 512-dimensional embedding via a fully connected layer. This 512-dim vector serves as the memory for the Transformer decoder.

B. Vision Transformer-Based Captioning Model

The ViT encoder is a pretrained ViT-B/16 [8] that divides each 224×224 input image into $14 \times 14 = 196$ non-overlapping 16×16 patches. Each patch is linearly embedded and processed by 12 Transformer encoder layers with hidden dimension 768 and 12 attention heads. The classification token (CLS) output, which aggregates global scene information, produces a 768-dimensional representation that is projected to 512 dimensions via a linear layer.

C. Shared Transformer Decoder

- Both models use an identical caption decoder comprising:
- A learned token embedding matrix of shape $|\mathcal{V}| \times 512$, where $|\mathcal{V}| = 2,004$.

- A learned positional encoding of shape 128×512 .
- A two-layer Transformer decoder with 8 attention heads and model dimension $d_{\text{model}} = 512$.
- A linear output projection from 512 to $|\mathcal{V}|$.

The decoder is trained with causal (autoregressive) masking and cross-attends to the encoder’s projected feature vector as the memory context.

IV. EXPERIMENTAL SETUP

A. Dataset

Both models are trained and evaluated on the Flickr30k dataset [13], which contains 31,014 images each annotated with five reference captions. The standard Karpathy split is used: 29,000 training images, 1,014 validation images, and 1,000 test images.

B. Vocabulary

A shared vocabulary is constructed from the training captions using word-level tokenization. The 2,000 most frequent words are retained, yielding a vocabulary of 2,004 tokens including four special tokens: $\langle \text{PAD} \rangle$ (index 0), $\langle \text{UNK} \rangle$ (index 1), $\langle \text{SOS} \rangle$ (index 2), and $\langle \text{EOS} \rangle$ (index 3).

C. Image Preprocessing

All images are resized to 224×224 pixels and normalized using ImageNet statistics (mean $[0.485, 0.456, 0.406]$, standard deviation $[0.229, 0.224, 0.225]$). No data augmentation is applied during training.

D. Training Configuration

Both models are optimized using Adam with differential learning rates and weight decay selected through a preliminary Optuna-based hyperparameter search. For each model, 15 tuning trials are run for 2 epochs per trial, searching over encoder learning rate, projection learning rate, decoder learning rate, dropout, and weight decay. The selected ResNet configuration is: dropout = 0.1498, encoder LR = 4.123×10^{-5} , projection LR = 2.6975×10^{-4} , decoder LR = 1.9844×10^{-4} , and weight decay = 4.208×10^{-7} . The selected ViT configuration is: dropout = 0.3234, encoder LR = 3.293×10^{-6} , projection LR = 6.261×10^{-5} , decoder LR = 2.4166×10^{-4} , and weight decay = 5.762×10^{-6} . The loss function is cross-entropy with padding positions ignored. Final training uses a batch size of 16, a maximum of 100 epochs, and early stopping with patience of 5 epochs and a minimum improvement threshold of 1×10^{-4} on the validation loss. All experiments are run on a Kaggle cloud instance equipped with a Tesla P100 16GB GPU (CUDA 12.8, PyTorch 2.10.0, Python 3.12).

V. EVALUATION METRICS

Caption quality is measured using four standard automatic metrics computed on the 1,000-image test set:

BLEU (Bilingual Evaluation Understudy) measures n-gram precision between generated and reference captions with a brevity penalty. We report BLEU-1 through BLEU-4 [14].

METEOR (Metric for Evaluation of Translation with Explicit ORdering) computes an F-score based on unigram alignment between hypothesis and reference, incorporating stemming and synonym matching [15].

ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation – Longest Common Subsequence) measures the longest common subsequence between generated and reference captions, rewarding fluency at the sentence level [16].

CIDEr (Consensus-based Image Description Evaluation) computes TF-IDF-weighted n-gram similarity between a generated caption and the set of reference captions, rewarding descriptions that are both accurate and consensus-aligned [17].

Inference latency is measured as the average wall-clock time per image on the test set using greedy decoding.

VI. EXPERIMENTAL RESULTS

A. Caption Quality

Table I reports caption quality scores on the 1,000 Flickr30k test images. The ViT-based model is slightly higher on BLEU-1 through BLEU-3, while the ResNet-50 model performs better on BLEU-4, METEOR, ROUGE-L, and CIDEr.

TABLE I
CAPTION QUALITY COMPARISON ON FLICKR30K TEST SET (1,000 IMAGES)

Metric	ResNet-50	ViT-B/16
BLEU-1	0.6512	0.6592
BLEU-2	0.4408	0.4450
BLEU-3	0.3009	0.3018
BLEU-4	0.2074	0.2045
METEOR	0.1993	0.1984
ROUGE-L	0.4951	0.4914
CIDEr	0.4263	0.4137

B. Computational Efficiency

Table II summarizes the computational characteristics of both models. The ResNet-50-based system converges significantly faster in both epochs and wall-clock time, while the ViT-B/16 model is substantially larger.

TABLE II
COMPUTATIONAL EFFICIENCY COMPARISON

Property	ResNet-50	ViT-B/16
Encoder weights	97.8 MB	330 MB
Training epochs	24	24
Best val. loss	2.4721	2.4163
Total training time	216.31 min	500.44 min
Avg. time / epoch	≈ 9.01 min	≈ 20.85 min
Avg. latency / image	41.06 ms	41.35 ms

C. Training Convergence

Both models begin with similar training losses (3.4188 for ResNet-50, 3.3862 for ViT-B/16 at epoch 1) and run for 24 epochs before early stopping. At epoch 24, the final training losses are 2.0400 (ResNet-50) and 2.1130 (ViT-B/16). The

ViT model reaches a lower best validation loss (2.4163 vs. 2.4721), but this does not translate into better corpus-level test metrics in this run. The ResNet-50 encoder also reaches its final model in substantially less wall-clock time (216.31 vs. 500.44 minutes).

D. Qualitative Examples

Both models generate grammatically correct and semantically relevant captions for most test images. Qualitative inspection indicates that the two systems are often similar on simple scenes with one dominant subject, while differences are more noticeable on crowded or interaction-heavy images. These observations are consistent with the mixed quantitative results, where neither encoder dominates every metric.

VII. DISCUSSION

A. Caption Quality

The updated results show a split behavior across metrics. ViT-B/16 yields slightly higher BLEU-1 to BLEU-3 scores, which suggests stronger short n-gram overlap with references. However, ResNet-50 performs better on BLEU-4 (+0.0029), METEOR (+0.0009), ROUGE-L (+0.0037), and CIDEr (+0.0126), indicating stronger higher-order phrase consistency and better consensus with human references in this experiment. Overall, the gains are modest but coherent enough to favor ResNet-50 on aggregate caption quality.

B. Computational Trade-offs

Even with tuned hyperparameters, the ViT encoder introduces substantial computational overhead. The ViT-B/16 pretrained weights are $3.4\times$ larger than ResNet-50 (330 MB vs. 97.8 MB), and the per-epoch training time is approximately $2.3\times$ longer (20.85 vs. 9.01 min), resulting in a total training time overhead of about $2.31\times$ (500.44 vs. 216.31 min). This difference is consistent with the higher computational burden of patch-level self-attention compared to convolutional feature extraction. Inference latency remains very similar for both models (41.35 vs. 41.06 ms per image), so the main practical cost difference appears during training rather than deployment-time inference.

C. Data Efficiency and Inductive Bias

Flickr30k provides only 29,000 training images, a moderate-scale dataset by modern standards. CNNs embed two forms of inductive bias—locality (each filter covers a limited receptive field) and translation equivariance—that allow them to learn effective visual representations from relatively few examples. ViTs lack these biases and instead learn global spatial relationships more directly from data, which often requires larger datasets or stronger regularization to fully surpass CNNs [8]. In this study, ViT-B/16 attains lower best validation loss, but ResNet-50 yields better BLEU-4/METEOR/ROUGE-L/CIDEr on the held-out test set. This suggests that on moderate-scale captioning data, CNN inductive biases still provide a meaningful generalization advantage.

D. Implications for Deployment

For applications where training efficiency and practical model size are important, ResNet-50 is the more favorable encoder in this setup: it trains over $2\times$ faster, uses much smaller encoder weights, and also achieves stronger BLEU-4, METEOR, ROUGE-L, and CIDEr. ViT-B/16 remains competitive and may still be attractive when short n-gram precision is prioritized or when scaling to much larger pretraining and finetuning datasets. In this experiment, however, the CNN-based encoder offers the stronger overall deployment trade-off.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a controlled comparative study of ResNet-50 and ViT-B/16 visual encoders within an identical encoder-decoder image captioning framework. Both models were trained and evaluated on Flickr30k under matched data and decoder conditions with tuned optimization settings. The ViT-B/16 encoder achieved slightly higher BLEU-1 to BLEU-3, but ResNet-50 achieved stronger BLEU-4 (0.2074 vs. 0.2045), METEOR (0.1993 vs. 0.1984), ROUGE-L (0.4951 vs. 0.4914), and CIDEr (0.4263 vs. 0.4137). ResNet-50 also reduced total training time by approximately $2.31\times$ (216.31 vs. 500.44 minutes) with a $3.4\times$ smaller encoder footprint, while inference latency remained comparable.

These results indicate that, for Flickr30k-scale captioning, CNN inductive biases still provide a strong quality-efficiency balance relative to ViT-based encoding, even when both systems are tuned and trained under a unified framework.

Future work will explore: (i) scaling to larger datasets such as MS COCO or CC3M to assess whether ViT’s quality advantage widens with more data; (ii) beam search decoding to improve caption fluency for both models; (iii) attention map visualization to interpret which image regions each encoder emphasizes; (iv) fine-tuning BLIP-2 [10] as a strong baseline for comparison; and (v) investigating encoder choices specifically for reducing hallucination, guided by evaluation protocols such as POPE [12].

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APPENDIX A

CODE LINK

https://github.com/PradipPokhrel55/caption_generator